Technology Stack (Architecture & Stack)

Date	12-06-2026
Team ID	
Project Name	**Voice Based Concept Understanding Analyser**
Maximum Marks	5 Marks

Technical Architecture:

Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Provides a simple and interactive web interface for students to record/upload audio, enter reference answers, view evaluation results, and download reports without	Streamlit
2.	Backend / Server-Side	Implements the core application logic, handles API requests, integrates AI models, processes user input, and coordinates communication between all system components.	Python 3.11 + FastAPI
3.	Speech-to-Text Engine	Converts students' spoken explanations into accurate text for further semantic analysis and evaluation.	OpenAI Whisper
4.	Semantic Understanding Engine	Computes semantic similarity between the student's explanation and the reference answer to assess	Sentence-BERT (SBERT)

		conceptual understanding instead of simple keyword matching	
5.	Generative AI Feedback	Generates personalized learning feedback, identifies missing concepts, suggests improvements, and provides detailed explanations based on the evaluation results	Google Gemini API
6.	Speech Analysis	Extracts speech features such as speaking rate, pauses, duration, and fluency metrics to evaluate communication quality.	Librosa
7.	Database / Data Storage	Stores user profiles, evaluation history, scores, reports, and system logs for future reference and analytics.	PostgreSQL
8.	Report Generation	Generates professional PDF evaluation reports containing scores, AI feedback, semantic analysis, and improvement recommendations.	ReportLab
9.	Version Control	Maintains source code, tracks project versions, enables collaboration, and simplifies deployment and maintenance	Git & GitHub.
10.	Deployment / Hosting	Supports local development and can be deployed online for remote access by students and faculty.	Local Environment / Streamlit Community Cloud (Future)